Laboratoire
d'ingénierie pour les
systems complexes

UNIVERSITÉ
Clermont
Auvergne

Forest scale planning For multifunctionality

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Importance of scale in forest ecology

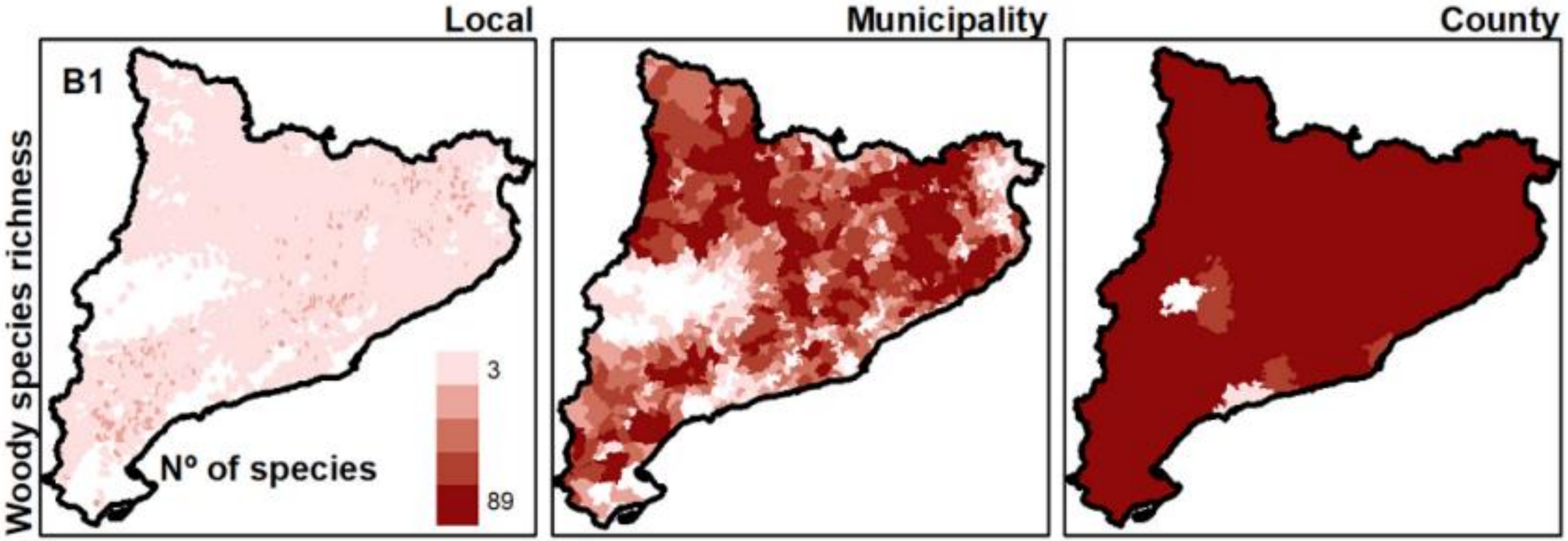

Figure 2: Spatial pattern of forest ecosystem services supply and their relationships Roces-Díaz et al. 2018

Forest planning scale

- **Current Planning Issues**
- Stand-level decisions dominate
- Highly fragmented ownership and planning
- Lack of coordinated management

- **But Ecosystem Processes Operate at Larger Scales**
- Windthrow vulnerability
- Pest outbreaks and wildfire spread
- Seed dispersal dynamics
- Habitat connectivity and biodiversity conservation

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=> Need for a multi-scale decision system to take into account both local and global constraints

Thank you for your attention

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Introduction

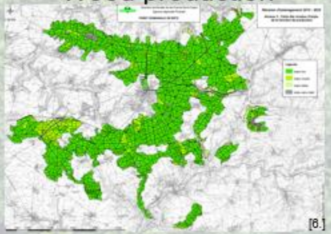
Managing forests effectively requires looking beyond individual forest stands and considering larger spatial and temporal scales. While stand-level approaches can have positive outcomes—such as boosting local biodiversity—they may be less effective at the landscape scale, where they can reduce habitat diversity through homogenization [1]. These include habitat loss, reduced diversity, and increased vulnerability to disturbances.

Landscape-scale planning should help balance different goals like timber production, carbon storage, and biodiversity conservation. Recent research suggests that diversifying forest management across space and time can support more resilient and multifunctional landscapes[2].

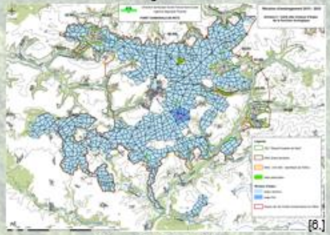
Because these challenges involve simultaneously deciding both what management actions to take and where to apply them, developing innovative methods is crucial to effectively balance multiple objectives and spatial complexities in landscape-scale forest management.

Forest functions

Wood production


[6.]

Biodiversity and habitat


[6.]

Social utility


[6.]

The maps reveal the spatial complexity of forest management challenges.

- Diverse functions and goals to address
- Uneven spatial scales: some very local, others global
- Overlapping zones, potential hotspots for conflict

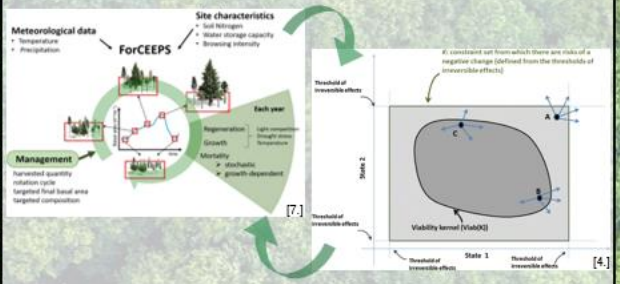
Method developpement

Challenges

The complexity of terrain and ecosystems across large landscapes presents significant challenges for management methods, including:

- Modeling ecosystems accurately at broad spatial scales
- Managing the computational complexity due to a large number of possible management actions
- Meeting multiple, often conflicting constraints simultaneously
- Adaptability of tools for different use (limit of optimality)

Goal



- Extend the mechanistic forest model ForCEEPS to predict dynamics at the forest scale
- Apply viability theory to formulate forest management as a control problem under multiple constraints

Conclusions

Managing forests at the landscape scale is challenging due to the complexity of ecosystems, the diversity of services they provide, and the need to balance production, biodiversity, social demands, and climate adaptation.

New tools like control theory can help by revealing system dynamics and offering flexible pathways, allowing managers to adapt strategies to their local context [4].

To make this possible, new governance structures will be needed—ones that support collaboration and long-term planning across scales [5].

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